

JULY 2023

OPERATING SYSTEMS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer Question No. 1 of Group A is compulsory.
Answer any Five (05) Questions from Group B.

Group A

1. Answer any five:

3 x 5=15

- i. Discuss Time sharing operating system.
- ii. What is the difference between logical address and physical address?
- iii. Write a short note on Thrashing.
- iv. What is external fragmentation? Explain with diagram.
- v. Discuss on different system threats.
- vi. Write different functions of operating system.
- vii. Explain First Fit with suitable example.

Group B

- 2. What do you mean by Batch operating system? How many page faults occur for the following reference string with four-page frames (4 PF) using – (i) FIFO and (ii) LRU Page Replacement algorithm: 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2, 0, 0. 3+3+3
- 3. State bounded buffer problem. Write critical section solution of Reader-writer problem. 3+6

Answer the following questions using the Banker's algorithm:

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				

- i) What is the content of the Need Matrix?
- ii) Is the system in a safe state? If 'yes' write down the safe sequence
- iii) If a request from process P1 arrives for (0, 4, 2, 0) can the request be granted immediately? Explain. 1+4+4
- 5. What is deadlock? What are necessary conditions to hold it? Explain Belady's anomaly with an example. 2+2+5
- 6. Read the following table and calculate Average waiting time in i) preemptive SJF scheduling and ii) FCFS scheduling 5+4

Process	Arrival time	Burst time
P1	0	8
P2	1	6
P3	2	4
P4	3	2

- 7. Suppose that a disk drive has 5000 cylinders, numbered from 0 to 4999. The drive is currently serving a request at cylinder 145, & previous request was at cylinder 125. The queue of pending request, in FIFO is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance covered by the following disk scheduling algorithms? -
 (i) SSTF, (ii) SCAN.
 Briefly explain Internal Fragmentation. 3+3+3

- 8. Write short note on i) Access Matrix ii) Cryptography iii) RAID 3+3+3